George Sharpe

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Education

University of Exeter — Exeter, UK
BSc in Computer Science, 2:1 — *2022 – 2025*

Sandringham School — St Albans, UK
A-Levels — *2020 – 2022*

- **Grades**: Computer Science (A*), Maths (A*), Further Maths (B) & Physics (B)

Work Experience

Canopius Group — Manchester, UK
Junior Data Engineer — *Sep 2025 – Present*

- Designed a novel algorithm in **T-SQL** independently to interpolate binder policy earned premiums, reducing inaccuracy from 23% to 0%. Validated results through **Excel**-based reconciliation. Used **SSMS**, **ADF**, **MDS**.
- Developed working knowledge of **Terraform** through cross-team collaboration and delivered initial infrastructure-as-code changes in production environments.
- Delivered BAU responsibilities by collaborating across colleagues and stakeholders, while quickly learning and applying internal systems such as **data warehouses** and **SQL Agent Jobs** ensuring timely resolution of requests.
- Proactively organised knowledge-sharing sessions to support junior colleagues, sharing technical expertise, system knowledge, and providing ongoing support within the team.

Sandringham School — St Albans, UK
Computer Science Tutor — *April 2021 – May 2021*

- Tutored students in programming concepts through coding challenges, debugging, and effective communication

Projects

- **Automated Clinical Coding - Python, Pandas, Transformers, Pytorch, Scikit-learn**:
  - Integrated clinical knowledge into LLaMA-3 prompts using a binary decision tree for ICD code predictions.
  - Built an end-to-end clinical NLP pipeline by integrating data from three MIMIC-IV subsets for hierarchical diagnosis classification.
- **Event Management Database - SQL**:
  - Designed ER diagrams and translated to normalised relational schemas with justified constraints.
  - Basic SQL queries using joins and subqueries.
- **Customer Segmentation for E-Commerce - Python, Pandas, Scikit-learn, Matplotlib**:
  - Applied K-Means clustering with optimal k via Elbow method; performed feature selection and scaling.
  - Visualised clusters and inertia curves to communicate insights effectively.
- **Atmospheric Advection Simulation - C, OpenMP**:
  - Implemented 2D finite difference solver with forward Euler stepping and modelled vertical wind shear.
  - Parallelised core computational loops with OpenMP for multi-core performance; generated output files.
- **Killer Sudoku Solver - Python, NumPy, SymPy**:
  - Represented the game state as a matrix. Used Gauss-Jordan elimination to produce a solution efficiently.

Skills

Languages: T-SQL, Python, Java, C#

Technologies: SSMS, ADF, MDS, SQL Agents Jobs, Terraform, Databricks, SSIS, Relational Databases, ETL Processes, Data Warehousing, Git